

TrialBridge

A computer helper that finds the right medical research study for a patient
- explained in plain language

First, what is a "clinical trial"?

Before a new medicine or treatment reaches every hospital, doctors test it carefully on volunteer patients. This test is called a **clinical trial**. Patients who join often get early access to new treatments, and their help makes medicine better for everyone.

But every trial has a strict list of rules about **who is allowed to join**. For example:

*"Must be 18 to 75 years old. Must have type 2 diabetes. Blood sugar (HbA1c) between 7% and 10.5%. Kidneys must be working reasonably well. **Not allowed** if you had a heart attack in the last 6 months, or if you are pregnant."*

The problem

There are **hundreds of thousands** of trials in the world, each with its own long list of rules, written in medical language. A busy doctor cannot read all of them for every patient. So most patients never hear about a trial that could have helped them, and many trials get delayed because they cannot find enough people. Today, this checking is mostly done **by hand**.

Our idea, in one line

TrialBridge reads a patient's medical details and the rules of thousands of trials, and suggests the trials that patient can actually join - and it explains why, rule by rule.

Think of it like a job portal. A job portal matches a person's CV to job requirements. TrialBridge matches a patient's health record to trial requirements. The difference is that a wrong match here matters much more, so our system must **show its reasons** instead of just saying "trust me".

It works **both ways**:

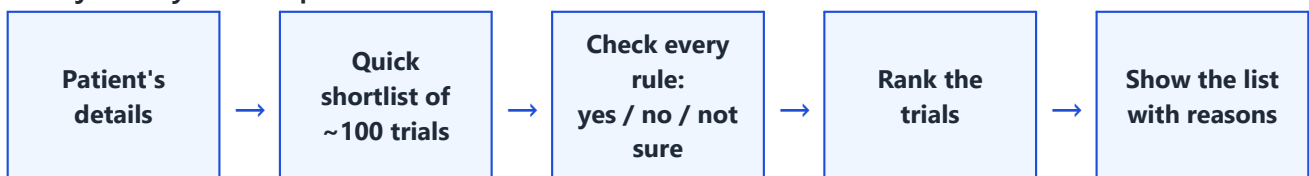
Who uses it	What they ask	What TrialBridge gives
A doctor / patient	"Which trials can this patient join?"	A ranked list of trials, each with a tick / cross / question mark next to every rule
A trial organiser	"Which of our patients fit this trial?"	A list of suitable patients from a group, best first

How it works - the 5 parts

Our group has 5 members, and the project is split into 5 parts (one per person). Here is what each part does, with an everyday comparison.

#	Part - what it does	Everyday comparison
1	Collect the trials Downloads trial details from the official worldwide list of trials (ClinicalTrials.gov) and stores them neatly.	<i>Like collecting every job advertisement into one folder.</i>
2	Understand the rules Reads each trial's rules written in English and turns them into simple checks a computer can do, e.g. "age between 18 and 75", "kidney test above 45", "no heart attack in last 6 months".	<i>Like turning a long paragraph into a checklist.</i>
3	Understand the patient Reads a patient's record (illnesses, medicines, test results) and arranges it in the same simple form.	<i>Like filling a patient's details into a standard form.</i>
4	Match and explain First quickly shortlists the trials that talk about the patient's illness (like a Google search). Then ticks off every rule on the checklist against the patient's form, and ranks the trials.	<i>Like a teacher checking an answer sheet against the answer key, marking each question.</i>
5	Test it and show it Measures how often our suggestions are correct, using an official "answer key" made by doctors. Also builds the website where you type a patient and see the results.	<i>Like a report card for our system, plus the shop window.</i>

The journey of one patient



Why "not sure" is important

Sometimes the record simply does not say something (for example, no recent kidney test). A careless system would guess. TrialBridge instead says "**not sure - please check**". Each trial is then labelled **Eligible**, **Possibly eligible** (some things need checking) or **Not eligible** (with the exact rule that failed). Being honest about what it doesn't know is safer for patients.

What makes our project different

Similar research projects mostly use a big AI chatbot (like ChatGPT) to read everything, which is slow, costly and can make things up. We do most of the work with **clear, checkable rules**, and plan to use AI only for the leftover rules that are too vague for simple checks. We also plan to add **India's own trial list (CTRI)**, which most such tools ignore.

The data we use (and why it is safe)

A computer system needs examples to work with. Here is what we use:

Data	What it is, simply	How much
ClinicalTrials.gov	The official, free, public website of the USA that lists trials from all over the world. This is our source of real trials.	2,766 open trials for the demo, plus 26,158 more for testing
Synthea (made-up patients)	A free tool that creates realistic but fake patients, complete with illnesses, medicines and test results. No real person's data is used , so there is no privacy risk.	300 fake adult patients
TREC 2021 "answer key"	A famous international test set. Doctors took 75 example patients and marked, for thousands of trials, whether each patient can join. We compare our answers against theirs - like checking exam answers against the official key.	75 patients, about 36,000 doctor-marked answers
n2c2 2018 (coming)	Another doctor-marked test set, rule by rule. Needs a free permission form, which we are applying for.	To be added in the next phase
CTRI (coming)	India's official trial list.	To be added in the next phase

What has been built so far

The first half of the project ("Phase 1") is **complete and working** on the computer:

Part	Status	What exists today
1. Collect trials	Done	Automatically downloads trials and saves them in a small database
2. Understand rules	Done	Turns rules into checks: ages, test values (with unit conversion), illnesses, medicines, "no history of...", "within the last 6 months"
3. Understand patient	Done	Reads the fake patients' records, or a patient described in plain English, e.g. "62-year-old man with diabetes, on metformin"
4. Match and explain	Done	Shortlists, checks every rule, ranks trials, and explains each decision
5. Test and show	Done	Scored against the doctors' answer key; website with 4 screens (below); 20 automatic self-checks all passing

The website has 4 screens

Screen	What you can do
Patient to Trials	Pick or type a patient, see the best trials, click "Why?" to see every rule ticked or crossed
Trial to Patients	Pick a trial, see which of the 300 patients fit it (takes about 1 second)
Rule Reader	Paste a trial's rules and watch the computer highlight what it understood
Results	Shows the report card: how often our suggestions were right

How good is it? (the report card)

We gave our system the 75 test patients and looked at its **top 10 suggested trials** for each. Then we counted how many of those 10 the doctors had marked as **"yes, this patient can join"**.

Method	Correct trials in the top 10 (on average)
Plain keyword search (like a basic search engine)	about 2.6 out of 10
Smarter search (understands medical meaning)	about 3.1 out of 10
TrialBridge (smart search + checking every rule)	about 4.5 out of 10

In simple words: **checking the rules made the suggestions about 43% better** - and this was done without any big AI chatbot. Also, the most relevant trial now appears noticeably closer to the top of the list.

Is 4.5 out of 10 low? No - this is a famously hard test, and many trials are "almost right" (the right illness, but the patient fails one rule), which even doctors must read carefully. Improving this further is exactly the goal of the next phase.

We also found that our rule reader fully understands about **1 in 4** trial rules and partly understands about **2 in 5**. The rest are vague, such as *"any condition that, in the doctor's opinion, makes the patient unsuitable"*. This tells us clearly where AI help is needed next.

What is remaining (Phase 2)

Still to build	In simple words
AI helper for vague rules	Use a free AI model running on our own computer (Llama / Mistral via "Ollama") only for rules our checklist cannot handle - and make it quote the patient record as proof, so it cannot make things up.
India's trial list (CTRI)	Add trials from India's official registry, so the tool is useful here.
Bigger medical dictionary	Right now it knows about 46 common illnesses and 17 medicine groups. Connect it to standard medical dictionaries (UMLS, SNOMED) so it knows thousands.
Second answer key (n2c2)	Test the system rule-by-rule on another doctor-marked set.
Fair comparison	Compare three versions side by side: rules only, AI only, and our mix of both.
Make it ready to share	Move to a proper database, package it so it runs on any computer, and put it online.

Who does what in our group

Member	Owns	In the next phase
Member 1	Collecting trials and storing data	Adds India's trial list (CTRI)
Member 2	Understanding the rules	Connects to big medical dictionaries
Member 3	Understanding the patient	Handles more kinds of patient records
Member 4	Matching and explaining	Adds the AI helper for vague rules
Member 5	Testing and the website	Runs the second answer key and the fair comparison

Small dictionary of words you may hear

Word	Meaning
Clinical trial	A careful, supervised test of a new treatment on volunteer patients
Eligibility criteria	The rules about who can and cannot join a trial
Inclusion / Exclusion	"You must have this" rules / "You must NOT have this" rules
Synthetic patient	A realistic, computer-made fake patient (protects privacy)
HbA1c, eGFR	Common blood tests: average blood sugar, and how well the kidneys work
Search / retrieval	Quickly finding the trials that talk about the patient's illness
Answer key (benchmark)	A test set marked by doctors, used to score our system fairly
LLM / AI model	A computer program that reads and writes text, like ChatGPT
Phase 1 / Phase 2	The first half of the project (done) / the second half (to do)

In short: clinical trials can help patients, but finding the right one is like finding a needle in a haystack. TrialBridge reads the trial rules and the patient's record, and acts as a fast, honest assistant that says *"these trials fit, and here is exactly why"*. The basic version already works and beats ordinary search clearly; next we teach it the harder rules and Indian trials.

Important: this is a student research project that helps people find options to discuss with their doctor. It does not replace a doctor's decision.